

Extraterrestrial infrastructures: Review of in-situ materials, envelope solutions, and energy performance of livable Martian and lunar habitats

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ABSTRACT

Long-term human exploration of Mars and the Moon, such as envisaged by the Artemis program, depends on developing sustainable, energy-efficient habitats. This review synthesizes current research on the challenges and strategies for constructing livable environments on these celestial bodies, with a focus on in-situ resource utilization (ISRU), building envelope design, and overall energy performance. It examines the physical and thermal properties of lunar and Martian regolith in relation to their potential as local building materials, considering a range of construction techniques for their feasibility and structural reliability. The analysis also covers energy management solutions, highlighting approaches to power generation and thermal regulation essential for long-term habitability. The findings emphasize the promise of ISRU as base materials, expanding the research scenario beyond traditional structural applications, while identifying ongoing challenges in thermal control as well as power generation and storage. The review concludes by outlining research gaps and recommending pathways to reduce the dependency on pre-fabrication and to minimize the necessity of resupplies from Earth, embracing the direction of sustainability that human space exploration aims to adopt.

Abbreviations and Symbols

AM	Additive Manufacturing	LAD	Lunar Astronomical Days
AMDROHP	Additively Manufactured Deployable Radiator with embedded Oscillating Heat Pipes	LAN	Lunar Astronomical Nights
ATCS	Active Thermal Control System	LE	Laboratory Experimentation
EATCS	External Active Thermal Control System	LE	Laboratory Experimentation
ECLSS	Environmental Control and Live Support System	OSM	On-Site Measurement
ECSS	European Cooperation for Space Standardization	PCS	Process Cooling System
EEATCS	Early External Active Thermal Control System	PPE	Power and Propulsion Element
EVA	Extra-Vehicular Activity	PTCS	Passive Thermal Control System
HRS	Heat Recovery System	PV	Photovoltaic
IATCS	Internal Active Thermal Control System	PVTCs	Photovoltaic Thermal Control System

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ISRU	In-Situ Resource Utilization	RFC	Regenerative Fuel Cells
ISS	International Space Station	SHC	Specific Heat Capacity
KPI	Key Performance Indicator	SP	Specific Power
LE*	Laboratory Experimentation, conducted on simulants	TELP	Thermal and Electrical Conductivity Probe
LEO	Low-Earth Orbit	TRL	Technology Readiness Level
LRO	Lunar Reconnaissance Orbiter	VHC	Volumetric Heat Capacity
MLI	Multi-Layer Insulation	λ	Thermal Conductivity
MMODs	Micro-Meteoroids Orbital Debris	ρ	Density
MMS	Mojave Mars Simulants	c	Specific Heat Capacity
OM	Orbital Mapping	c_v	Volumetric Heat Capacity

1. Introduction

The Moon and Mars are now the focus of research on sustainable

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space architecture, resource use, and energy-efficient habitat design as we aim for long-term extraterrestrial habitation. The ability to build habitable spaces that can tolerate drastic temperature changes, high radiation levels, and the logistical challenges of transferring materials from Earth is essential to the success of such missions. In-situ resource utilization (ISRU), which makes use of locally accessible materials, has become a crucial tactic for the building of interplanetary infrastructure due to the prohibitive cost and complexity of space cargo transfer.

The International Space Station (ISS) and the future Lunar Gateway [1] are two examples of existing space habitats that offer important insights into pressurized living conditions, energy management, and thermal control. Permanent construction on planetary surfaces, however, presents new difficulties, including the requirement for self-sufficient energy sources, long-term exposure to space weathering, and structural integrity under reduced gravity and potential seismic forces (e.g., moonquakes). In particular, the Martian and lunar regolith offer promising potential for 3D printing and construction, reducing dependence on Earth-based materials.

The material qualities, building methods, and energy systems required for viable Martian and lunar habitats are reviewed in this work under three main themes.

1. In-situ materials and construction techniques: Evaluating the thermophysical characteristics and composition.
2. Designing the building envelope: Assessing radiation shielding techniques, thermal performance, and structural arrangements.
3. Energy management and sustainability: Examining temperature control systems, energy end-uses, and power-producing options.

This assessment attempts to identify knowledge gaps and suggest future strategies for improving the viability of extraterrestrial habitats by combining previous developments and ongoing research. The results will aid in the engineering and design of self-sustaining, energy-efficient structures that could sustain human life beyond Earth. To help the reader better understand the structure of this paper, Fig. 1 provides a holistic overview of the content.

2. Environment and in-situ resource utilization (Isru) on The Moon and Mars

2.1. Atmospheric and climatic conditions

The thin atmosphere of Mars, which is primarily made up of carbon dioxide (~95,3 %), nitrogen (~2,7 %), and argon (~1,6 %) [2], causes notable temperature fluctuations at the surface. The average low temperature is -153°C , while the average high temperature is 20°C . While the air temperature at head height stays about 0°C , the surface temperature at noon at the equator can reach 24°C [3]. These extreme fluctuations pose challenges for habitat insulation and temperature regulation. With carbon dioxide offering horizontal shielding of more than 100 g/cm^2 and vertical protection of $16\text{--}22\text{ g/cm}^2$, Mars' atmosphere provides some natural radiation shielding [4]. However, radiation exposure remains a constraint that needs to be addressed during design due to the lack of full protection that can cause harm to human organisms, which are used to the amount filtered by the terrestrial magnetic field.

The Moon, by contrast, has almost no atmosphere, leading to extreme temperature differences between sunlit and shadowed regions. With an atmospheric mass of only $\sim 25 \times 10^3\text{ kg}$ and a surface pressure of $3 \times 10^{-15}\text{ bar}$ at night, it is essentially a vacuum [5]. Due to its low gravity ($\sim 1/6$ of Earth's), gases cannot accumulate at the surface. At the lunar equator, daytime temperatures reach 121°C , while nighttime temperatures plummet to -133°C [6]. These conditions require advanced thermal control systems to sustain human life.

The Moon's axial tilt of $\sim 5,14^{\circ}$ to the Ecliptic leads to permanent irradiance in certain polar regions, and permanent shadowing the local craters. Because of its nearly constant solar exposure, which is essential for energy production, the lunar south pole in particular has been recognized as a desirable site for human habitation [7]. The Moon's surface is also extremely exposed to solar and cosmic radiation since it lacks a strong magnetic field and atmosphere. This calls for strong habitat shielding techniques such as using regolith-based protective layers or underground structures. A regolith layer approximately $2\text{--}3\text{ m}$ thick offers radiation shielding equivalent to about 40 g/cm^2 . For

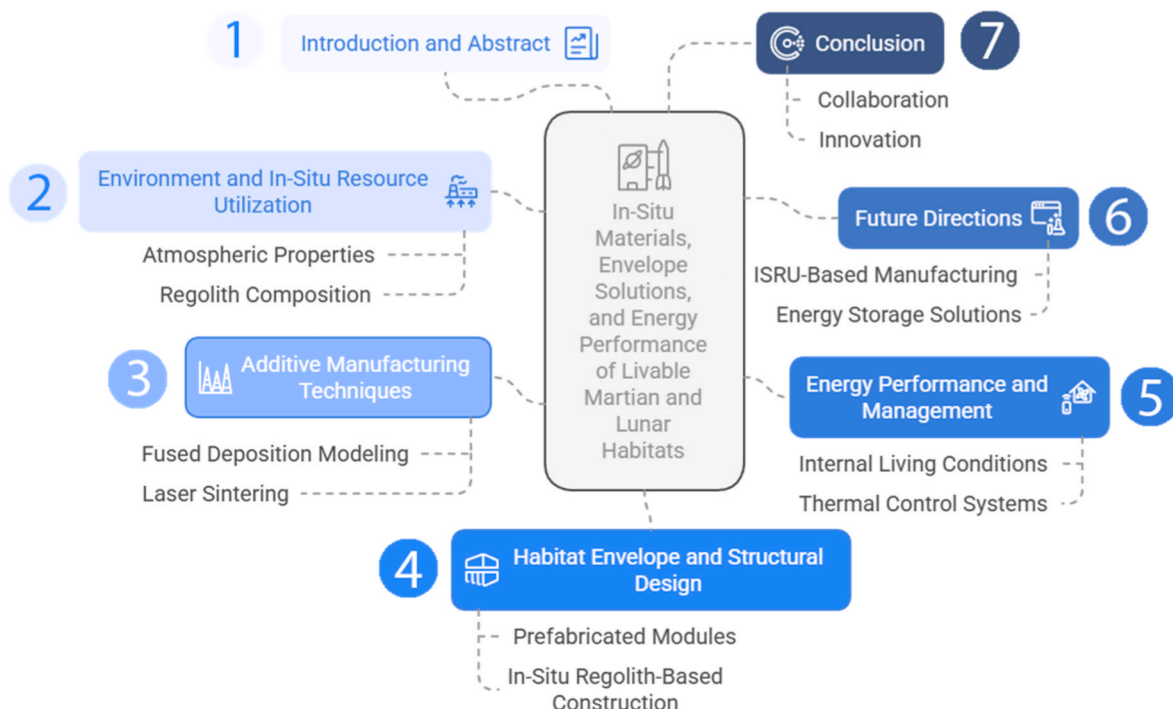


Fig. 1. Visual overview of the study.

comparison, Earth's atmosphere provides roughly 100 g/cm^2 of protection. Interestingly, at just 1 m beneath the lunar surface, radiation exposure drops to levels similar to those on Earth ($\sim 2 \text{ mSv/year}$) [6]. Additionally, at low gravity, electrostatically charged lunar regolith sticks to surfaces, presenting astronauts with health and mechanical difficulties. These factors must be accounted for in habitat design and astronaut safety protocols.

For the purpose of this publication, other relevant environmental factors characterizing the geophysics and the atmosphere of the celestial bodies have not been taken into account, privileging the fundamental constraints for the energy management of an ET habitat. Those constraints are summarized in Table 1, that lists the atmospheric characteristics of the Moon and Mars compared with Earth.

2.2. In-situ resources and materials

A critical aspect of extraterrestrial construction is the availability and suitability of in-situ resources. Mars' surface consists of a heterogeneous, unconsolidated soil layer primarily composed of silicates and basaltic materials [10]. This presents an opportunity for utilizing local resources in construction. Regolith-based concrete is among the most studied materials in the context of ET habitats structural design. Basaltic fibers exhibit elasticity comparable to aluminum, with a breaking stress level superior to certain plastics, such as low- and high-density polyethylene, utilized on Earth as structural fiber reinforcements. Notably, basalt also offers radiation protection [11]. The main knowledge derives from spectrometric measurements as well as from samples analyzed by rover missions landed on the planet in past decades showing that the chemical compositions of collected quantities include Silicone Dioxide (SiO_2 , ~ 43 up to 46%), Iron Oxide (FeO , ~ 16 up to 21%), Aluminum Oxide (Al_2O_3 , ~ 9 up to 10%), Magnesium Oxide (MgO , ~ 6 up to $8,6\%$), Calcium Oxide and Sulfur Trioxide (CaO and SO_2 , both 6 up to 7%) and others ($\sim 5\%$) [11]. Given its basaltic composition, Martian regolith has structural properties similar to some Earth-based construction materials [12]. Fig. 2 reports the chemical compositions of several Martian samples, as well as Lunar, in comparison to Earth soil.

Regarding the thermophysical properties of Martian soil, unlike the case of the Moon, investigations into Martian soil have primarily relied on simulants derived from basalt-based materials, informed by data collected from landed probes and remote observations. Table 2 reports the values described in the publications born after these studies. Johansen (1975) [13] estimated Thermal Conductivity (λ) between $0,5$ and $1,64 \text{ W/m-K}$ according to the quantity of ice, while Nagihara et al. (2023) [14] conducted experiments on Mojave Mars Simulants (MMS), providing insights into their thermal and physical properties. Different density mixtures were analyzed, providing insights for λ , which vary from $\sim 0,05$ to $0,06 \text{ W/m-K}$, and Specific Heat Capacity (c) between ~ 615 and 650 J/kg-K . Siegler et al. (2012) [15] studied the properties change of regolith according to the ice content, with a mixture created by vapor deposition of water below the triple point, at low atmospheric pressure, simulating the frost land at Mars' poles: λ increases from $0,075$

to 1009 W/m-K as the 45% -porosity sample gets filled by ice particles, with an increase of density (ρ) from a standard 1570 kg/m^3 to a remarkable 2488 kg/m^3 . These variations highlight the importance of site-specific material characterization before processing.

Data from the Phoenix (2008) and InSight (2018) missions support these value trends. Based on orbital observations from the Thermal and Electrical Conductivity Probe (TELP), Zent et al. [17] studied a mixture composed of $\sim 60\%$ basalt and a variable quantity of hematite (iron oxide), observing that as the hematite content dropped from 40% to zero, both thermal conductivity (average $\sim 0,085 \text{ W/m-K}$) and density (average $\sim 1600 \text{ kg/m}^3$) also decreased. Early results from the InSight mission, described by Grott et al. (2021) [16], showed even lower values - thermal conductivity around $0,039 \text{ W/m-K}$, specific heat capacity at 630 J/kg-K , and density at 1030 kg/m^3 . Overall, the data suggest that compression significantly affects density, which can vary from about 1000 to 1770 kg/m^3 , while temperature influences specific heat. For instance, Nagihara et al. (2023) [14] found SHC values between 650 and 750 J/kg-K across temperatures from 180 K to 260 K .

Like Mars, several methods can be used to approach constructing lunar habitats, and much effort has been put into studying lunar soil characteristics for future processing activities. Unlike Mars, the Moon's surface can rely on samples that have been returned to Earth and studied for over 50 years, from which simulants have been developed for further investigation. Due to the features of the corresponding soil, the Lunar surface can be briefly divided into the Lunar *Maria* and *Highlands*. Stoesser et al. (2010) furtherly divide the lunar Highlands according to the categories of High-Ti Mare and Low-Ti Mare [18]. Most property models tend to follow this classification, with chemical weights that vary according to the area of origin. The chemical composition of lunar soil is mainly constituted by Oxygen ($\sim 41\text{--}42\%$), Silicon ($\sim 20\text{--}21\%$), Iron ($\sim 12\%$), Calcium ($\sim 8\%$), Aluminum ($\sim 7\%$), Magnesium ($\sim 6\%$) and others (4%) [19,20]. Heiken et al. provided a summary of mineral proportions according to Apollo and Luna sampling sites [21], characterizing more than 90% of lunar rocks, silicates are the most common minerals in the lunar crust. Similarly to Mars, important quantities of basalt are present. The Moon's surface contains significant amounts of oxides, constituting approximately 40% of the overall composition, primarily iron oxides and volcanic glass, which are rich in oxygen. Lunar soil samples have also revealed small but notable traces of native metals like nickel and copper [22]. Finally, recent studies confirmed the presence of water bodies in the form of ice mixed with soil and rocks, in correspondence with the permanently shadowed region of craters' bottoms. Limited quantities have also been observed in the form of volatiles ($\text{OH}/\text{H}_2\text{O}$). Even if the utilization of these resources is a promising element for future missions' success on the satellite, their accessibility is still an object of study [23,24].

The lunar materials' thermophysical features based on samples and simulants are summarized in Table 3. With density of granular soil depending on the depth, Xiao et al. (2022) [25], Hayne et al. (2017) [26], Vasavada et al. (2017) [27] and Morgan et al. (2018) [28] studied the properties change of the soil according to the compaction of the sample, describing how density (ρ) vary from $\sim 1800 \text{ kg/m}^3$ in a range between 60 and 100 cm depth, to 1100 kg/m^3 at the surface. Very low density, approximately 470 kg/m^3 , is noticed for samples collected in the proximity of craters' rims, suggesting that the soil in these areas may have been turned over by impact events [22]. All the sources report a linear proportional relationship between density and thermal conductivity, varying from $0,0015$ to $0,0120 \text{ W/m-K}$, when considered within the specific temperature ranges and conditions of external pressure in which the studies were conducted. These conditions have been highlighted in the right column of Tables 2 and 3 Yu et al. (2014) [29] reported the result of the orbital mapping activity of the Lunar Reconnaissance Orbiter (LRO), highlighting a difference between the density (ρ) and specific heat capacity (c) of Maria's regolith and Highlands', which is generally higher.

Specific heat capacity is a fundamental property that also determines

Table 1
Summary of Earth, Moon, and Mars' atmospheric properties, from [2,8,9].

Property	UM	Earth	Moon	Mars
Surface area	km^2	$510,1 \times 10^6$	$37,9 \times 10^6$	$144,8 \times 10^6$
Radius	km	~ 6.371	~ 1.739	~ 3.396
Mass	kg	5972×10^{24}	7342×10^{22}	$6,418 \times 10^{23}$
Gravity at the equator	m/s^2	9,81	1,62	3,71
Period of rotation	h	24	656	24,62
Temperature range	$^\circ\text{C}$	Max: +58 Min: -89	Max: +121 Min: -133	Max: +24 Min: -153
Length day	Earth days	1	29,5	1,02
Surface atmospheric pressure	kPa	101,325	3×10^{-13}	6×10^{-3}

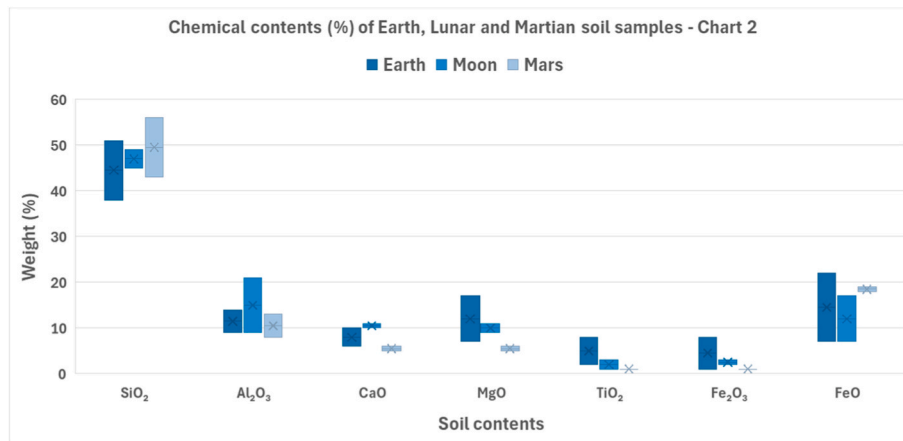


Fig. 2. Chemical weights of Earth, Lunar, and Martian soil samples.

Table 2
Summary of λ , c , c_v , ρ values for Martian samples.

Reference	Study Technique	λ W/ m·K	c J/ kg·K	c_v J/m ³ ·K	ρ kg/ m ³	Study Condition
Nagihara, 2023 [14]	LE*	0,0520	650	0,8x10 ⁻⁶	1230	At different densities, vacuum
Nagihara, 2023 [14]	LE*	0,0560	617	0,95x10 ⁻⁶	1540	At different densities, vacuum
Nagihara, 2023 [14]	LE*	0,0600	633	1,05x10 ⁻⁶	1660	At different densities, vacuum
Grott, 2021 [16]	OSM	0,0390	630	0,69x10 ⁻⁶	1095	Collected on Surface
Zent, 2009 [17]	OSM	0,1400	597	1,05x10 ⁻⁶	1760	High (~40 %) hematite mixing
Zent, 2009 [17]	OSM	0,0700	721	1,05x10 ⁻⁶	1456	Poor (~10 %) hematite mixing
Siegler, 2012 [15]	LE*	0,0752	771	1,21x10 ⁻⁶	1570	No ice content
Siegler, 2012 [15]	LE*	1,0092	342	0,85x10 ⁻⁶	2488	Full ice content
Johansen, 1975 [13]	LE*	0,0500	–	–	–	No ice content
Johansen, 1975 [13]	LE*	1,5400	–	–	–	Full ice content

the amount of energy or power required for regolith processing, and calculation models are different depending on the temperature. In general, specific heat values increase proportionally up to approximately 900 J/kg·K as the temperature rises to around 350 K. Beyond that point, the rate of increase slows, continuing more gradually up to about 1500 K. At this point, values remain steady at ~ around 1500 J/kg·K [35,32]. For molten samples, ρ and c values tend to increase as the temperature rises, passing from ~2700 to 2900 kg/m³ for Highlands samples and from 2500 to 2700 kg/m³ for Maria's. Schreiner et al. (2015) [35] provided a series of analytical models to describe λ

variations of different solids to temperature variation. Models show proportionality between the numbers until the threshold of 1200 K, at 0.1 W/m·K, from which λ sharply increases up to >1 W/m·K. Reached the peak at 1500 K, values decrease below 0.1 W/m·K. Further, Woods-Robinson et al. (2019) [30] demonstrated the inefficiency of recent models for the determination of λ at lower temperatures (<50 K), proposing a new one and estimating thermal conductivity at 20 K as equal to 0,000275 W/m·K, showing therefore a great decrease. Values reported in sources align with results from the Apollo 12 and 14 missions [33]. The most recent Chang E-6 mission returned samples from the Moon's far side in the Fall of 2024, highlighting density values of ~3035 kg/m³ [34], while thermal properties are still in a study phase. Conductivity values tend to differ according to the sample's density, temperature conditions, and, therefore, the location and depth where the samples were collected. According to the processing activity to take place, granular and mineralogical conditions of regolith need to be chosen carefully [18,35].

Regarding utilizing in-situ lunar regolith for construction 3D printing, there are concerns about the regolith's stability against external pressures. The lack of a practical experience represent an uncertainty for additive manufacturing in ET environment. Radiation can cause microcracking in the structure, vacuum stability or outgassing, and humidity gradients within the habitat can cause damage to certain materials [16]. There are also concerns with thermal stresses due to extreme temperature differences caused by sun angle, lack of sunlight, and diurnal thermal cycling [18]. However, the magnesium found in lunar regolith is abundant and can be stronger than other proposed materials for habitat construction and should be also utilized for radiation shielding [17] with a recommended thickness of ~3.5 m of regolith to shield the habitat [19].

Future studies on lunar and Martian regolith could explore additional properties such as porosity, optical behavior, and albedo. For habitat design, a deeper understanding of mechanical properties will also be essential. Fig. 3 provides a comprehensive overview of the order of magnitude of λ , c , and ρ values of lunar and Martian samples presented in Tables 2 and 3 and described in Section 2.2.

3. Habitat envelope and structural design considerations

Building envelope design is essential for long-term Moon and Mars missions. Habitats may be fully or partially prefabricated on Earth or built onsite with in-situ resources. KPIs (Key Performance Indicators) for habitat evaluation could include payload, volume capacity, sustainability, and cost, as key factors. Leading examples with high Technology Readiness Level (TRL) include ESA's Lunar Habitation (with Foster + Partners) and the Moon Village (with SOM and MIT), both using or suggesting prefabricated structures and lunar regolith elements

Table 3Summary of λ , c , ρ values for lunar samples.

Reference	Study Technique	λ	c	ρ	Study Condition	
		W/m-K	J/kg-K	kg/m3		
Xiao, 2022 [22]	OSM	0,0015	680	0,32x10 ⁶	471	Collected on Surface
Xiao, 2022 [22]	OSM	0,0085	680	0,12x10 ⁶	1883	~1 m depth
Hayne, 2017 [26]	OM	0,0340	700	0,77x10 ⁶	1100	Collected on Surface
Hayne, 2017 [26]	OM	0,0074	428	0,77x10 ⁶	1800	~1 m depth
Yu, 2014 [29]	OM	0,0012	418	0,77x10 ⁶	1843	Observed in Highlands
Yu, 2014 [29]	OM	0,0012	358	0,77x10 ⁶	2150	Observed in Maria
Woods-Robinson, 2019 [30]	LE*	0,0012	4500	5,8x10 ⁶	1300	High Temp. (300–500K) ranges
Woods-Robinson, 2019 [30]	LE*	0,0003	500	0,65x10 ⁶	1300	Low Temp. (20K) ranges
Vasavada, 2012 [27]	LE	0,0006	–	–	1100	Collected in Surface, vacuum
Vasavada, 2012 [27]	LE	0,0070	–	–	1800	~1 m depth, vacuum
Schreiner, 2015 [31]	LE	1,0633	1600	44,0x10 ⁶	2750	Molten Regolith (1500 K), vacuum
Schreiner, 2015 [31]	LE	0,0039	248	3,7x10 ⁶	1500	Temp. 350 K ranges, vacuum
Schreiner, 2015 [31]	LE	0,0016	273	4,09x10 ⁶	1500	Temp. 100 K ranges, vacuum
Morgan, 2018 [28]	LE + LE*+OM	0,0120	630	0,49x10 ⁶	1500	Collected on Surface
Morgan, 2018 [28]	LE + LE*+OM	0,0120	630	0,94x10 ⁶	1790	~50 cm depth
Wakabayashi, 2006 [32]	LE	0,0133	506	7,8x10 ⁶	1550	Temp. 200 K ranges
Wakabayashi, 2006 [32]	LE	0,0221	891	13,8x10 ⁶	1550	Temp. 500 K ranges
Hemingway, 1973 [33]	OSM	0,0010	273	4,54x10 ⁶	1664	Temp. 100 K ranges
Hemingway, 1973 [33]	OSM	0,0014	253	4,55x10 ⁶	1800	Temp. 100 K ranges
Li, 2024 [34]	LE	–	–	–	3035	Collected on the Farside

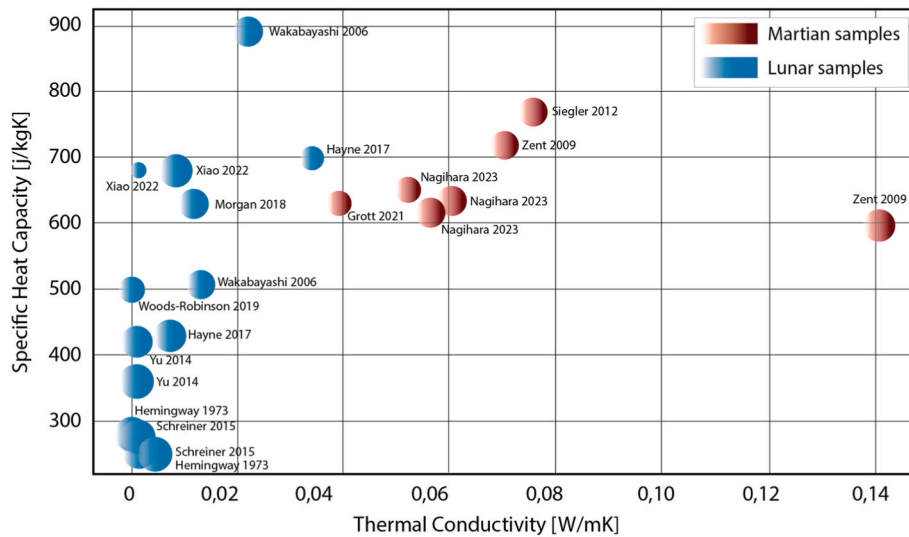


Fig. 3. Summary of λ , c , ρ values for lunar and Martian samples, represented by blue and red dots, respectively. Thermal Conductivity (W/m-K) is on the X-Axis, while Specific Heat Capacity (J/kg-K) is on the Y-Axis. The diameter of the datapoint is proportional to the density (kg/m³) of the samples. Ref. to Tables 2 and 3 (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

[36–38]. Fig. 4 represents a conceptual diagram highlighting the relations between the described KPIs with the habitat classes.

Prefabricated rigid habitats, in missions like ISS, Salyut, Skylab, Mir, and Gateway, utilize ring frames, beams, and pressure shells for structural and pressurization load [1,39,40]. Usually, a primary structure composed of ring frames, beams, longerons, and pressure shells provides structural integrity, bearing principal loads [39]. A secondary structure of rigid and semi-rigid shells then transfers internal loads caused by internal pressurization. Materials evolved from steel to aluminum and carbon fiber, often layered for thermal, pressure, and radiation protection, with added MMOD shielding [6]. Fig. 5 reports the broken down sections of the Moon Village envelope: with focuses on the stratigraphy and materials utilized, every layer has been designed to address an environmental constraint, as reported in the CFD Report [36].

The same principle applies to inflatable structures. Various concepts for inflatable and deployable modules have been explored since the 1960s, beginning with TransHab. The considerable advantage of this class of solutions compared to the rigid ones lies in the higher available

controlled space to parity of launch mass. For example, the BEAM module illustrates their efficiency in expanding internal volume while reducing launch weight [41,42]. Inflatable habitats, explored since the 1960s, use strong, lightweight materials (Kevlar, Vectran, Spectra) and rigid frames, but are more vulnerable to leaks and mechanical impacts [43]. MLI technology well conjugates with expandable shells, providing highly efficient insulating solutions, and their application is at the frontier of surface and orbital habitation [44], having being adopted in several experimental studies as main Passive Thermal Control System (PTCS) [45–47]. Fig. 6 shows an example of how inflatable materials can constitute an effective envelope for human living habitats: structural elements are on the external side, to absorb MMOD impact and contain internal overpressure, while the other layers are following towards the interior, separated by retention liners.

ISRU-derived construction focuses on using local materials like regolith to minimize launch payload and boost sustainability [48]. Methods include regolith shielding, 3D printing with basalt fibers, or making polymers for radiation protection [49] and window elements

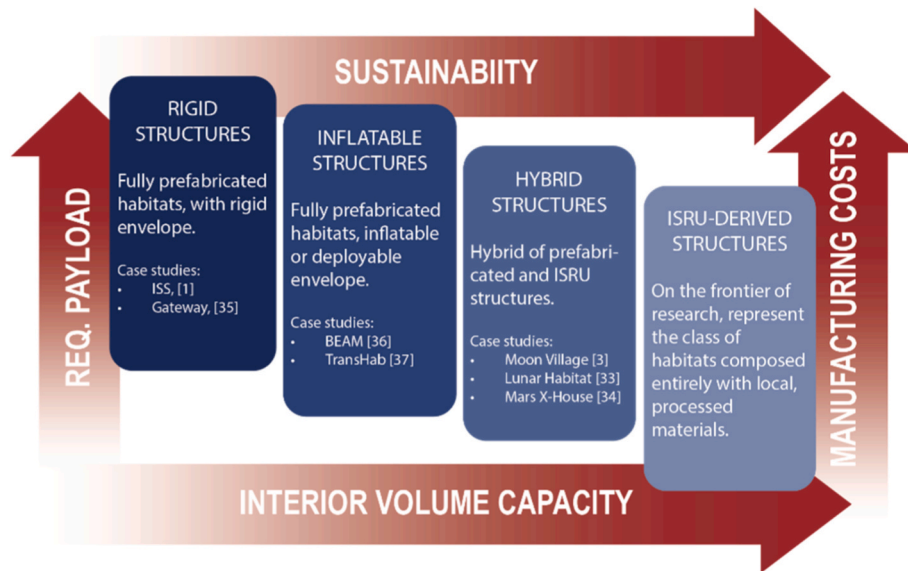


Fig. 4. Extraterrestrial habitats classification, with characterization in terms of payload, interior volume, cost, and sustainability.

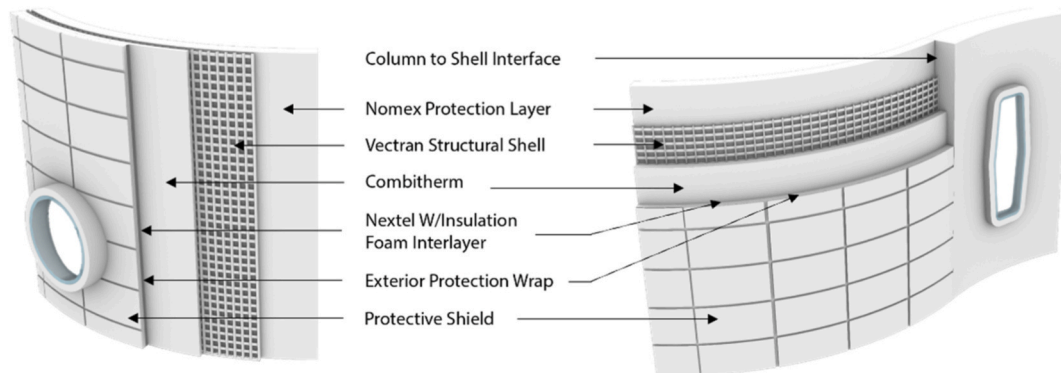


Fig. 5. Moon Village envelope concept details; credits to ESA, SOM [7].

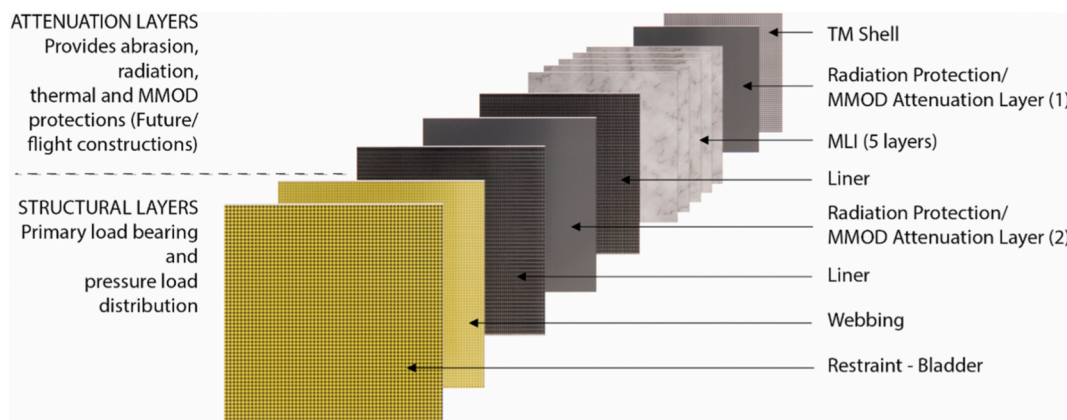


Fig. 6. Inflatable envelope components; credits to Hinkle, 2008 [47].

[3], up to research that open to the possibility of obtaining propellant from the soil [50,51]. At the moment of writing this paper (Spring 2025), there are no case studies foreseen to be built uniquely with local resources, without involving prefabricated components. It is possible for in situ 3D printing to be utilized in combination with other methods of habitat construction to create internal structure and separation within the building [45]. Fig. 7 provides an example of concept design focused

on In-Situ Resources Utilization: like described in the previous image, also in this project the external 3D-printed layer represents the shell structure, imagined in regolith concrete with basalt fiber reinforcement, also AM-realized; insulation layers are instead conceived in pre-fabricated materials.

From the construction point of view, a variety of Additive Manufacturing (AM) techniques are currently available and under study,

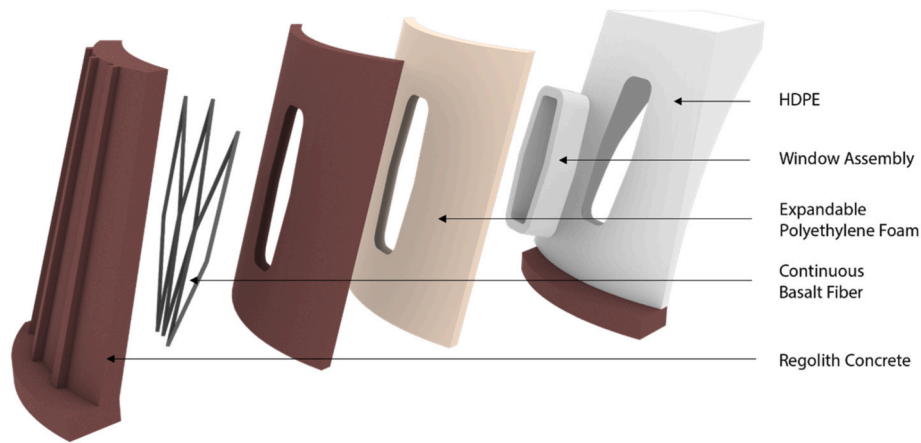


Fig. 7. Layered wall composition Mars X-House; credits to Yashar et al., 2019 [3].

differentiating by the materials availability, shapes and layouts, and solidification of the final product [52]. One of the reviewed methodologies include Fused Deposition Modeling (FDM), based on semi-solid material depositing in the x-y direction and temperature control for foreseen viscosity [53]. On the other hand, several techniques do not foresee the any addition or extrusion, acting instead directly on the preparation materials bed. For example Stereolithography, consists of polymerization and solidification through a beam of light operating on a resin bath, allows very high resolution [54]. Similarly, Balla et al. (2012) [55] suggested the possibility of using lenses to convey laser beams directly on the just-deposited material to maximize the sintering performance of the process. Furthermore, Laser Sintering, based on a powder depositing and solidifying process, is one of the most studied methodologies and can be applied to the processing of a wider spectrum of materials, including plastic, ceramics, glass, and metals [56,57]. Sintering the regolith transforms the powdered material into a solid mass through heating but not liquefying. In this case, the materials reach a temperature close to the melting point, allowing particles to couple. However, not all can solidify with the unique supply of a light beam, and according to the binding method and the utilization or not of a chemical binder, they can be categorized as follows: **Solid State Sintering**, **Chemically Induced**, **Liquid Phase Sintering**, **Partial Melting**, and **Full Melting** [58]. All of them achieve high mechanical properties without requirements for structural support. Among the others, a notable one to mention is the RegoLight project, pursued by the German Aerospace Centre - DLR and others, that aimed to realize structural bricks from pure lunar regolith, without chemical and other binding additives [59]. Another technique is Material Jetting, which is similar to Stereolithography and uses photopolymers as the raw material. The resin is applied in a single deposit, which reduces material consumption but usually requires structural supports during printing. Finally, Binding Jetting consists of the deposition of a binding additive on the raw materials, similar to what is done with sintering but without the activating light beam [60]. Mainly utilized on glass and sand, it requires post-process coatings to reduce the presence of printing waste particles. A wide research application is currently being pursued in the field, testing chemical addition to increase overall performance according to the green body shape [61–63]. Table 4 reports the energy consumption of each technique in an extraterrestrial environment, as listed by Isachenkov et al. (2021) [58], it can be highlighted how sintering techniques are the most energy-consuming, while deposition/jetting tends to be less demanding, and therefore may be preferred.

Other techniques are currently under evaluation for application in harsh environments, with contexts characterized by vacuum, low or no gravity conditions, and sharp temperature changes, which can heavily affect the manufacturing process. Further investigation directions go from the exploitation of water resources for structural and radiation-

Table 4
Summary of applicable AM techniques for Lunar and Martian missions.

Reference	Technique	Post Treatment	Energy Consumption [58]
Cano-Vincent 2021 [53]	Fused Deposition Modeling	–	Low (only moving)
Wallace 2014 [54]	Stereolithography	Sinterization, cleaning, and polishing	~50 kWh
Balla (2012) [55]	Lens Sintering	Cleaning, polishing	~2,3 MWh/m ³
Kruth 2005 [56]	Sintering or Selective Sintering	–	Up to a few MWh/m ³
Wang 2002 [57]		–	
Howe (2017) [64]	Selective Microwave Sintering	–	~600 kWh/m ³
Williams (2015) [62]	Material Jetting	Cleaning, polishing	Low (only moving)
Perry (2014) [60]	Binding Jetting	Cleaning, polishing	Low (only moving)

shielding purposes to energy system components for heat management [65]. Finally, in recent years, we have witnessed an increasing concern regarding the psychological health of crewmembers, most of all regarding the utilization of heavily shielded, enclosed regolith-based structures, and the prolonged isolation required for long-duration space missions. Developing solutions to avoid the bunker effect is in progress; however, while this is beneficial to the quality of life in the habitat, they often fail to meet the required standards for radiation shielding, and physical protection, as their resistance to tension and compression is not equalized by a sufficient capacity to bear to micro-meteoroids impacts. At the same time just a few polymeric materials, such as for example Borated Polyethylene, have demonstrated relevant shielding capacities, and none of them tested in LEO environment [3,46,66].

4. Energy performance and management in extraterrestrial habitats

4.1. Indoor living conditions requirements

NASA intends for a permanent base on Mars to eventually sustain a crew of 20 people for about 6–12 months. It should have areas for living, working, service, medical, and a greenhouse [43]. The characteristics of the interior environment of extraterrestrial habitats progressively changed, with the first experimentations in the contexts of the Apollo

Program and Skylab Program, which adopted 34,5 kPa, 100 % Oxygen, and 34,5 kPa, 74 % Oxygen, respectively [67,68]. In the last years, a lower limit was established to 48 kPa total pressure [69]. NASA Technical Standards and Reports provide guidelines for habitat internal environmental conditions, highlighting the importance of the mission as the main design driver. Habitats designed for a mission with high-frequency Extra-Vehicular Activity (EVA) phases should operate at 57 kPa total pressure, 34 % Oxygen; on the contrary, a habitat designed to allow limited EVA (Extra-Vehicular Activity) phases should maintain 101 kPa, 21 % Oxygen; finally an intermediate capacity should operate at 70,7 kPa and 26,5 % Oxygen content, to prevent damages to the lungs and allow pre-breathe protocols before EVA [70]. On the ISS, a slight overpressure of 101,3 kPa, a constant temperature of 22°C, and stable humidity (~60 %) characterize the pressurized indoor air condition. Oxygen presence in the indoor environment is the same as on Earth [71]. Air leakage from spacecraft or modules is an important consideration in atmosphere design. Ground testing has established realistic leakage rates of approximately 0,01 kg/day per module [67]. The standard aligns with ISS conditions, further providing indications regarding the presence of diluent gas in the internal atmosphere (at least 30 %), heat storage required between 4,7 kJ/kg of a crewmember, a heat loss of –4, 1 kJ/kg of the crewmember and atmospheric contaminant matter concentration to less than 3 mg/m³ with a crew generation rate of 1,33 mg/person-minute [72].

Future designs for beyond-Earth habitats must account for potential environmental requirements in areas designated for experimental activities. This includes ensuring that any deviations from specified conditions do not compromise the integrity or fairness of the experimental processes being conducted. To support the design, a wide literature of standards and guidelines is accessible for further reference, beyond the experimentation conducted on space stations and vehicles. NASA Technical Standards System (NTSS) provides an extensive frame of indications [73] as example, the “NASA Spaceflight Human System Standard Volume 2 - Human Factors, Habitability, and Environmental Health” [74] illustrates Agency-wide requirements that minimize health and performance risks for flight crew in human spaceflight programs, while the “Human Integration Design Processes (HIDP) handbook [75] illustrates the planning and the execution steps to address those targets. Through the European Cooperation for Space Standardization (ECSS), ESA provides aligned documentation: the “ECSS-E-ST-10-04C Rev.1 - Space environment” [76] defines the characteristics of the internal ambient of spacecraft and other vehicles operating in space regime, providing general models and guidelines to meet local induced environment targets. In a future direction, the “ECSS-U-ST-20C - Space sustainability - Planetary protection” [77] describes specifics, requirements, and protocols for robotic and human missions beyond Earth’s celestial bodies.

4.2. Energy end-uses in ET habitats

Given the challenges of designing space habitats for the Moon and Mars, energy demand can be broadly divided into a few main categories of loads. The first and most complex to quantify are the energy loads driven by mission-specific objectives, as they are highly dependent on the mission’s planning and scope. These include activities such as field research and extravehicular activities (EVAs), which also encompass the recharging of indoor vehicles; the collection, analysis, and processing of local materials for manufacturing purposes, including tasks related to habitat expansion and construction; efforts aimed at achieving long-term mission sustainability, such as the operation and maintenance of on-site food production systems like greenhouses; and the powering of indoor technical equipment, including computing devices and processing tools. They may also require dedicated thermal regulation systems, such as Process Cooling Systems (PCS) and Heat Recovery Systems (HRS), operating in separate loops as seen in the Internal Active Thermal Control System (IATCS) of the International Space Station (ISS). The

second category involves energy loads derived from the crew’s biological activity. These are influenced by factors such as body surface area, metabolic rate, and diet, and vary according to the specific characteristics of the mission.

The third category pertains to energy demands associated with Energy Storage Systems (ESS), which are essential when the habitat relies entirely on solar photovoltaic (PV) arrays, as is the case with the ISS. Electrochemical battery systems are the most used power storage technologies currently available, with a strong literature and a wide range of experience and tested practices. The lifespan of these systems is given by the number of discharge cycles and their cycle efficiency, which determines energy losses during charging and discharging. However, the primary factor guiding their selection is specific energy, measured in Wh/kg [78]. On the Moon, where day-night cycles last an average of 29.5 Earth days, the PV system must not only power the habitat during periods of illumination but also charge the ESS to sustain operations during the prolonged lunar night. The performance of these systems depends on the energy storage technology employed. While Lithium-ion (Li-ion) batteries are prevalent on Earth, hydrogen regenerative fuel cells (RFC), even currently in a low TRL stage, may offer a more viable solution in space, particularly if local resources can be utilized, due to the higher power densities compared to Li-ion devices [79]. To take into account the results of the design of the ESS, integrated in a well-conducted management of power loads, it is necessary to adopt power adjustment coefficients [79,80]. Although a full review of energy storage systems (ESSs) for space missions is beyond the scope of this work and would require a dedicated paper, current applications mainly rely on lithium-ion batteries. Future developments are expected to include solid-state batteries, supercapacitors, and other emerging technologies [81].

Finally, the combination of the three load types, together with the energy needed to maintain stable indoor conditions, defines the design of the Environmental Control and Life Support System (ECLSS). This system that typically includes air, water, and waste management represents the final major category of energy loads that must be integrated into the habitat’s overall energy strategy [67].

The available practices are the ones planned or in operation in Low Earth Orbit (LEO) or around the Moon. The ISS, as it orbits Earth, allows a crew of about 3–13 people to live and work in space. It has a volume of 900 cubic meters and a mass of 400.000 kg. It generates about 735.000 kW-hours of electricity per year from 2247 square meters of solar arrays [1,2]. A different approach characterizes the Lunar Gateway: conceived not with the target of permanent missions in orbit, but as a bridge supporting human crewed activities and future settlements on the lunar surface [5], the station has significantly reduced sizes (with 63.000 kg it has about 1/6 of ISS’s mass), it is designed to support 2 up to 4 people for a 30-day mission, and its Power and Propulsion Element (PPE), once fully deployed, will generate up to 60 kW of power [5].

HI-SEAS (Hawai’i Space Exploration Analog Experiment), LUNARES, Colozza (2020), and the Moon Village are examples of research that provided valuable insight into the energy management discipline in extraterrestrial surface habitats design. These terrestrial analogs, simulate and analyze the habitat and mission concepts, listing the energy control, generation, and storage systems, making assumptions regarding the operation loads could help with sizing these systems. These key hypotheses are reported in Table 5.

Taken together, these studies and analogs underline how energy demand in space habitats is shaped by multiple, scenario-dependent factors, from thermal losses and crew loads to equipment provision and power supply choices.

Each source emphasized different aspects in their studies: Engler et al. (2019) and Kaczmarzyk et al. (2020) focused on sustaining human life by testing crew survival in extreme environments [79,82], Colozza (2020) studied 3 separate configurations of power generation and storage [83], and the Moon Village was proposed as a permanent settlement on the lunar surface [36]. Regarding heat losses through the

Table 5

Comparison for Energy Management Study Assumptions; credits to Engler (2019), Kaczmarzyk (2020), Colozza (2020), ESA (2020) [82], [79], [83], [36].

Category	Assumption	Source 1 [82],	Source 2 [79],	Source 3 [83],	Source 4 [36],
Mission	Mission scope	Human life sustains	Human life sustains, exploration	ISRU O ₂ production	Exploration, long-term settlement
Mission	Mission location	Martian surface	Lunar surface	Lunar surface	Lunar surface
Mission	Mission duration	Up to 365 days	14 days	1 lunar day-night cycle (708 h)	300 days
Mission	Crew size	6 persons	6 persons	6 persons	4 persons
Mission	Habitat's volume	~368,1 m ³	~412,6 m ³	~267,9 m ³	~698,3 m ³
General	Key parameters for energy modeling	Intravehicular physical activity profile The mental health of crew members Power demand for electrical devices ESS type	Crew-derived heat loads Intravehicular physical activity profile Heat losses through the envelope Power demand for electrical devices ESS type	3 different power gen. + ESS study cases Demanding an ISRU manufacturing plant Constant solar radiation Heat loss through beyond-envelope components Power demand for electrical devices	Transfer to the Moon scenario Redundancy concepts Heat losses through the envelope Power demand for electrical devices ESS type
PTCS	Envelope concept	Opaque;	Opaque rigid structure; MLI shell. Trans.:15,47 W/m ² [84].	Opaque inflatable structure; MLI shell. 4,09	Opaque hybrid structure; double MLI inflatable shell, ISRU shell at the base. 17,6
PTCS	Envelope heat losses (kW)	–	–	–	–
ATCS	Equipment heat load	0,81 kW	1185 W	7 (eq.) + 25,8 (ISRU O ₂ plant)	57,4 kW
ATCS	Crew-derived heat load (kW)	1,81	0,113	–	0,5 - 2
ATCS	ECLSS heat load (kW)	0,11	6,59	21,05 kW (heater excl.)	9,2
Power	Power system	Photovoltaics (100 %), propane generator (backup)	Photovoltaics (100 %)	Photovoltaics (cases 1, 2); nuclear fission (case 3)	Photovoltaics (59,1 %), nuclear fission (40,9 %)
Power	Power generation (kWe)	10,0	25,95	53,85	144,0
Power	ESS system	Li-ion battery system	Li-ion or RFC battery system	Li-ion batteries (case 1); RFC (case 2);	RFC battery system
Power	ESS electrical load (kWe)	28,50	Up to 10,63	28,05	59,84

envelope, Kaczmarzyk et al. (2020) relied on academic background assuming ~15 W of heat lost per sqm of shell [84]. Colozza (2020) simulated the dissipation through a highly-insulated envelope designed as a series of MLI shells considering an assumed-constant solar radiation of 1,35 kW/m², the contribute of regolith's albedo and a surface temperature of 102 K (~–171°C) [83]. ESA (2020) also calculated a 17,6 kW loss from the internal 20°C environment to the 65 K (~–208°C) temperature surface during the lunar night through an envelope with a similar design [36]. For crew-derived heat loads, Engler et al. (2019) focused on the occupants well-being as the primary driver for energy engineering design, identifying patterns between mental health and energy consumption [82], while Kaczmarzyk et al. (2020) surveyed metabolic rates and physical activities of the LUNARES analog missions' participants, calculating the sensible and latent heat load produced by crew members [79].

These research projects highlight the strong dependence of energy systems on the specific mission scenario, as they are heavily influenced by the equipment installed in the habitat. For instance, while the values reported in Table 5 based on Engler et al. (2019) represent averages across different HI-SEAS mission scenarios [82], Kaczmarzyk et al. (2020) calculated energy demand assuming full equipment provision, including the consumption of EVA vehicles for surface exploration [79]. Further, the team working on the Moon Village imagined as a permanent human settlement on the Moon, provided a much higher heat load from equipment and tools, five times the amount reported in the analogs [36]. Colozza (2020) also provided a valuable analysis of structuring habitat tools and proposed an O₂ production ISRU-based plant that manufactures oxygen and hydrogen via regolith heating and subsequent electrolysis, assumed active during the lunar day [83].

Regarding power, with the second and the third sources assuming 100 % PV generation, with electrical energy directly converted to heat for loads calculation, ESA (2020) is the first study to consider a mixed photovoltaic and nuclear generation, taking into account the volatility

of PV systems during lunar nighttime frames. The calculations regarding power generation capacity and power storage capacity are produced summarizing the demand for electrical devices and ECLS systems, along with the demand for full battery charging, with 20 % of the safety margin considered by ESA (2020) in their calculations [36]. Colozza (2020) finally proposed three different case studies, simulating the scenario of the O₂ production plant, the scenario with exclusively the habitat equipment, and the scenario with the two features combined. The generation and storage systems differ, proposing an integration of photovoltaics and nuclear fission with Li-ion batteries (case 1) and RFC (case 2) [83]. Kaczmarzyk et al. (2020) considers only nuclear generation, without dependency on PV and therefore without ESS [79]. The sources provide indications regarding arrays and reactor sizes, along with specific energy and dimensions provisions for ESS, with the final aim of calculating the expected launch payload and therefore the mission's cost demand for energy management systems.

5. Thermal and power systems

Future research on extraterrestrial built structures' energy and sustainability targets two main areas: Thermal Control Systems and the systems of Energy Sources and Generation that enable them. This chapter provides brief introductions to both.

5.1. Thermal control: active and passive systems

As an example of thermal control in space environments, space stations, such as ISS, rely on a combination of different contemporary working devices and methods to regulate their internal climate. These systems can be broadly classified into two categories: Active Thermal Control Systems (ATCS) and Passive Thermal Control Systems (PTCS) [85]. The latter includes systems that do not actively control cooling but are designed to continuously dissipate heat through their material

properties. This is primarily achieved through reflective paneling, exterior materials, and insulation techniques. From the energy management point of view, Multi-Layer Insulation (MLI) is a critical layer, which is a thermal insulation composed of multiple layers of thin sheets that overlap to provide high insulation performance by utilizing the vacuum between highly reflective layers, distanced by low-conductive spacers. Due to necessity to exclude heat convection in the vacuum chambers, MLI, typically only a few millimeters thick, performs effectively under extremely low-pressure conditions, preventing air infiltrations from jeopardize the insulation process. It acts as a flexible blanket that conforms to the rigid structure it insulates. The insulation is made up of alternating highly reflective materials, such as metallized films, and low-conductivity materials. Fiber spacers separate the layers, maintaining a vacuum chamber that enhances insulation. Common materials used in MLI include Kevlar, Nextel, Nomex, Twaron, and Aracron, which provide a combination of strength and thermal efficiency [86,87]. Another complementary technology is the Thermal Radiators. Being the ambient normally characterized by vacuum, conduction, and convection, are not viable heat removal and irradiance processes is the preferred means of dissipation. Made of lightweight materials to facilitate transportation and deployment, components are characterized by high-emissivity coatings to enhance radiative efficiency and are usually placed in areas less exposed to direct sunlight to maximize heat rejection. For example, a new deployable radiator system is under development through a collaboration between JPL and academia: at the heart of this innovation lies the Additively Manufactured Deployable Radiator with embedded Oscillating Heat Pipes (AMDROHP), a technology composed of an evaporator and a condenser connected by a thermal switch, designed to efficiently transfer heat across dynamic or moving interfaces [88]. As depicted in the following table, there are further methods of small-scale passive thermal control that designers use on CubeSats and SmallSats (small spacecraft): some of these act as a barrier avoiding heat from transferring through the envelope, such as MLI, sprayable liquid coatings applicable films, sun shields, and phase change materials; other equipment are utilized to dissipate heat, directing it through the components and eliminating the excess, such as thermal louvers, thermal switches, and conductive joints [89,90]. The technologies are summarized in the following Table 6.

Utilized when Passive control is not sufficient for load supply, ATCS has three main purposes: heat collection, heat transportation, and heat rejection. To achieve this, on the ISS, four systems are utilized: IATCS, External Active Thermal Control System (EATCS), Photovoltaic Thermal Control System (PVTCS), and Early External Active Thermal Control System (EEATCS). While the frontier of active technologies is summarized in Table 7, Fig. 8 provides a schematic of part of the ATCS on the ISS, highlighting the relations among the different subsystems. The IATCS circulates water through the module's interior to collect excess heat from electronic devices and experimental equipment, then distributes it to the Interface Heat Exchangers to transfer to EACTS. The EEATCS is a temporary system used to collect, transport, and reject waste heat from habitable areas on the ISS, and it utilizes ammonia and radiators to expend heat into space [91]. The EACTS collect heat from the Interface Heat Exchangers and external electronic equipment and eventually transport it to radiators to reject into space. Due to the vacuum conditions and the lack of a heat transfer medium, irradiance happens to be the only possible way for heat release in the surrounding environment. The PVTCS also uses ammonia loops but collects excess heat from the Electrical Power System and eventually transfers it to PV radiators to dispose of it into space.

The harsh extraterrestrial environment demands careful control of large thermal loads and, where possible, continuous sensible heat production. As radiators continue to be the only viable means of heat rejection in the absence of an atmosphere, they necessarily must be oversized. Another potential technique is high-capacity thermal storage, with released heat from storage during shadow hours. The use of solar radiation for immediate thermal generation—comparable to solar

Table 6

State of the art of PTCS technologies for small spacecraft, re-elaborated from Weston et al. (2025) [90].

PTCS technology	Type	Description
Thermal Control Coatings	Envelope	Specialized liquid coatings are applied to spacecraft surfaces to regulate the balance between absorptivity and emissivity, ensuring optimal thermal performance and energy efficiency.
Films, Tapes	Envelope	Materials used to modify the surface properties of the spacecraft, aiding in temperature regulation and thermal control.
Multi-Layer Insulation	Envelope	Typically applied to the spacecraft's exterior, this material minimizes thermal conductivity and manages radiation exchange.
Thermal Straps	Envelope	Flexible conductive links that transfer heat from a source to a thermal sink via conduction.
Thermal Conductive Joints	Equipment	Interfaces that enable heat transfer through direct contact between two pressed surfaces; performance can be enhanced using interface fillers or conductive gaskets.
Sunshields	Equipment	Deployable structures are designed to reduce incoming solar radiation and limit thermal load on the spacecraft.
Thermal Louvers	Equipment	Equipment items that modulate heat transfer between spacecraft surfaces by adjusting surface exposure.
Thermal Radiators	Equipment	Dedicated surfaces that dissipate excess heat into space through radiative transfer.
Phase Change Materials	Equipment	Substances enclosed in metal compartments absorb thermal energy by changing phase when exposed to heat.
Thermal Switches	Equipment	Electromechanical devices that regulate electrical current flow in response to temperature variations.

Table 7

State of the art of ATCS technologies for small spacecraft, re-elaborated from Weston et al. (2025), [77].

ATCS technology	Type	Description
Electrical Heaters	Equipment	Flexible space heaters that use an electric resistance element sandwiched between two insulating sheets to generate heat.
Cryocoolers	Equipment	Compact refrigeration systems are designed to achieve extremely low temperatures, ideal for cooling sensitive spacecraft components.
Thermoelectric Coolers	Equipment	Solid-state devices provide localized cooling by leveraging the Peltier effect, where an electric current passing through two different metals removes heat.
Heat Exchangers	Equipment	Devices are designed to mitigate excess sensible heat in the spacecraft modules or excess condensation water, meeting humidity targets.
Fluid Loops	Distribution	A system that circulates a working fluid throughout the spacecraft, transferring heat to designated sinks for temperature regulation.

thermal panels in ground systems—is similarly highly unrealized. Finally, advanced habitat structures require redundancy and flexibility to cope with various module requirements, operating profiles, and emergencies [92]. Advanced Thermal Control (ATC) systems are then designed with modular and independent subsystems, as evident in the Gateway Integrated ECLSS Model (GIEM) [93].

5.2. Power systems

Energy generation plays a pivotal role in the sustainability of long-term missions on Mars and the Moon, with photovoltaics at the forefront of experimentation, driving research toward pioneering uses even for terrestrial applications [94]. In this section, the major drivers

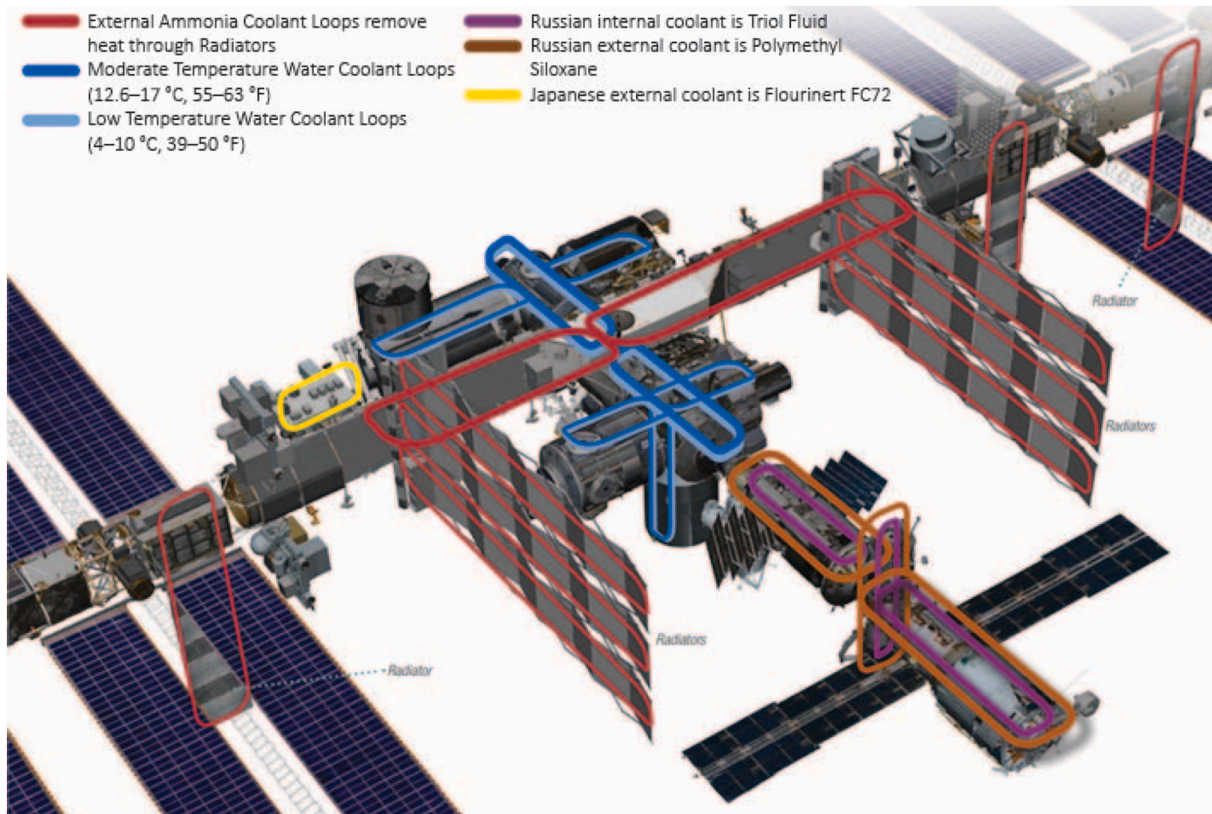


Fig. 8. Schematic diagram of the Active Thermal Control System (ATCS) for ISS, credits to NASA, 2015 [91].

regarding power generation is discussed.

5.2.1. Solar generation

To date, human spaceflight operations in Earth orbit and cislunar space have relied almost entirely on photovoltaic (PV) systems. The International Space Station is providing up to ~ 85 kW at top power, and the Lunar Gateway is designed for a ~ 60 kW capacity. Solar technology has improved considerably in the last decades, with the development of more efficient layer systems. Commercial products composed of monocrystalline, high-efficiency layers reach a value of 22 % efficiency, while experimental multi-junction cells overcame the value of 47 % in 2025 [95]. SmallSats and CubeSats commonly utilize high-performance solar cells with efficiencies exceeding 32 %, despite their significantly higher cost compared to terrestrial silicon solar cells, which achieve around 20 % efficiency [96]. Technologies that are currently at the frontier of research is based on multi-junction cells, organic solar cells, fuel cells,

and flexible solar cells [90].

Due to the inability to generate power during the planetary or Lunar nights and the degradation over the mission duration (caused by aging and radiation), solar cell usage is strongly limited by reduced efficiency in deep-space environments. Even with 68 % efficient tripe-junction cells (theoretically estimated by De Vos (1980) [97]), using solar technology has limitations beyond earth, also their performance decline as the distance from the sun increases. Fig. 9, re-elaborated from Weston et al. (2025), describes the phenomenon comparing the performances of solar arrays, differentiated according to the SP; further, it illustrates the areas where 100 m^2 solar arrays could operate, and their decreasing efficiency. Consequently, this constraint heavily affects the size of the PV system to be foreseen for human missions beyond Earth. The substantial energy required for daily human activities in a habitat may demand a large surface area of solar panels, determining a great payload during launching phases. Currently, spacecraft missions tend to fall within

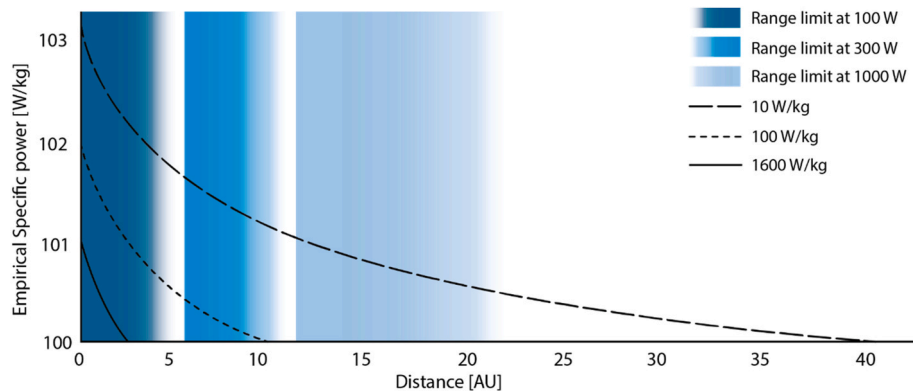


Fig. 9. Power operational limit for spacecraft, showing performance (lines) and operational areas (colored regions) of solar arrays. Credits to Weston et al., 2025 [89].

Specific Power (SP) of ~ 30 W/kg [90]. The opportunity provided by in-situ manufacturing may open higher mass arrays replicating Earth-fabricated cells for larger settlements on Mars and the Moon.

The utilization of PV and solar thermal systems as the main energy generation system is also conditioned by the morphology that characterizes the Moon and Mars: photovoltaic generation may not be practical for habitat locations where sunlight is scarce, such as settlements in craters or underground areas and require installation away from the colony and transmit. Astronomical and atmospheric conditions also play an important role. In addition, meteoroid impacts and the accumulation of lunar dust pose significant risks to solar array performance and longevity. For the Moon, which represents the most studied case, the state of permanent equinox determines the daytime duration on the lunar surface at the equator to be equal to the night time. Lunar astronomical day (LAD), defined as the time when the Sun is above the line of the horizon, are therefore if lunar astronomical nights (LAN), defined as the time when the Sun is below the same line [5]. Power generation investigation will therefore be affected by loads and selenographic coordinates. As examples of different approaches, Colozza et al. (2020) calculated the power production with an iterative process on MatLab, as a function of TCS [98], while Halbach et al. (2021) provided a design of vertical PV arrays exploiting the permanent illumination conditions at the Shackleton Crater, in the context of ESA's Moon Village project [99]. However, the energy request load for a solar generation system may be affected by different power management strategies, due to the system's dependency on the lunar day-night cycles.

5.2.2. Nuclear fission

As an alternative to PV, nuclear fission has started to be investigated for powering extraterrestrial human habitats since Project Horizon, developed at the end of the 50s in the context of the Cold War [100]. At the moment of writing this article (Spring 2025), NASA is planning to install a nuclear power generation device by 2030, while China and Russia made a similar announcement in May 2025, targeting the end of the 2035 [101]. There are several reasons behind this choice, with mass and in-situ conditions as key factors influencing the performance and impacting the reliability of the mission power generation, most of all in deep space scenarios, where PV technology utilization happens to be not feasible. In contrast, nuclear energy requires far less effort to produce the vast amount of thermal energy needed, making nuclear technology particularly attractive for this purpose. Novel technologies embraced in the family of Small Modular Reactors (SMR) happen to be likable candidates also due to the reduced mass payload that they would represent in the transportation to the Moon [101]. For example, NASA's Kilopower project consists of a 100 kW compact device where highly enriched uranium is supplied, generating heat that is then moved thanks to isolated pipes and sodium as heat transfer fluid in advanced Stirling engines that produce electricity. The waste heat is then transferred to the radiators or HRS [102,103]. Future mission designs that may adopt this approach will have, however, to consider constraints derived from radiation exposure, which may be severe and require dedicated shielding. AM technologies could also provide support in developing such structures, providing mechanical protection against MMOD impacts as well.

6. Future directions and recommendations

For future human habitation beyond earth, one of the major limitations remain to be the mass payload available for our current launcher systems. It has been emphasized in particular how solar energy generation is strongly dependent on the array's surface area and performance under dust and the risk of meteoroids, how fully prefabricated envelope solutions would require important efforts in terms of shielding construction and how thermal control depends on specialized devices built from materials with the right thermal properties. ISRU is surely part of a sustainable direction that the international community wants to give to space exploration but also determined by a very strict room to play in

terms of available mass to deploy from Earth. AM is one of the promising technologies in addressing these issues.

Additionally, the unique environmental conditions, including extreme temperature fluctuations, vacuum conditions, and radiation impacts, challenge the solutions designed to meet settling needs. Therefore, a clear understanding of the climatic surroundings in which a mission will operate is critical in establishing requirements for development. While broad consensus exists that the lunar South Pole is the most intriguing near-term option for settlement, substantial uncertainty now exists regarding the optimal location for durable human presence on Mars. Future and ongoing Artemis missions should reduce such uncertainty by testing feasible solutions and establishing habitats with the potential for long-duration habitation. A more comprehensive weather data and climate models are critical prior to colonization of such celestial bodies.

In the last year, we witnessed a boost in research focused on the sustainability of human life in space. The matter has been approached by several disciplines, each contributing from different perspectives, such as mobile habitats, designed for reduced or longer distance explorations on Mars and Moon Surfaces [104,105], floating habitats, designed to be suspended in the Earth-like gravity environment in Venus' atmosphere [106], underground habitats which foreseen in natural lava tubes and caves exploit the closed environment to protect from radiations; expandable habitats, such as the one proposed by Mangan (1988), consisting of a series of modular structures in defined geometries to be utilized as base building blocks [107,108]. These strongly interdisciplinary experiments will contribute to the concept of the first human settlement installed on a world beyond Earth.

Looking forward, advancing extraterrestrial infrastructure requires a multidisciplinary research agenda that goes beyond engineering efficiency to also consider human well-being, sustainability, and mission resilience. Future work should prioritize.

- Site-specific regolith characterization under operational conditions.
- Development of low-energy additive manufacturing methods suited for reduced gravity and extreme thermal cycling.
- Exploration of integrated power generation concepts, including ISRU-enabled systems.
- Habitat design strategies that address both technical protection needs and psychological health of crews.

7. Conclusions

Long-term Moon and Mars habitability depends on successful integration of in-situ resource utilization (ISRU), well-designed habitat envelopes, and efficient energy management systems. Much has been highlighted in this review regarding great leaps that have been made in the knowledge of regolith properties, construction techniques, and power generation technology but technical and scientific hurdles remain before the realization of fully self-sustaining extraterrestrial infrastructure.

The main insights are as follows.

- Lunar and Martian regolith in-situ resource materials possess high structural use and radiation protection capabilities but variable thermophysical characteristics with depth, mineralogy, and environment. More in-situ tests are required to reduce uncertainties.
- Habitat envelope can benefit from hybrid solutions, with prefabricated units supplemented by regolith-based shielding or 3D printing, to achieve payload efficiency with safety and performance requirements.
- Energy management will be a critical bottleneck. Even though photovoltaic is possible in the near term, future missions will need to pair them with advanced energy storage and nuclear fission reactors to overcome long lunar nights and Martian dust storms.

- Thermal control systems must address extreme diurnal fluctuations and vacuum. Redundancy, modularity, and innovative solutions (e. g., deployable radiators, regolith-based insulation) are necessary to maintain habitability.

Sustained human habitation of the Moon and Mars will be achieved only by interdisciplinary collaboration that brings together materials science, energy systems engineering, space architecture, and human factors. By introducing a framework for the challenges and opportunities among these fields, this review establishes a foundation for further research toward self-sufficient, extraterrestrial habitats.

CRedit authorship contribution statement

Cosimo Razeto: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rina Visvanathan:** Writing – original draft, Conceptualization. **Ehsan Kamel:** Writing – review & editing, Validation, Supervision, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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